

Bathala Slot Analysis

1. DESIGN & DEVELOPMENT PROCESS

Lucky888 was developed as a Bathala **simulation twin** for mathematical analysis and game design evaluation. The prototype reconstructs the reference game's observable 6×5-reel structure, symbol set, tumble mechanic, Bathala elimination skill, Scatter behaviour, Free Game behaviour, and multiplier system. Gameplay rules that can be directly verified through observation have been reproduced as closely as possible. However, as the underlying probabilities and gameplay parameters cannot be directly verified, the simulation twin uses configurable datasets (JSON & CSV files) that can be easily imported, adjusted, and exported out via the HTML5-based workbench.

DESIGN METHODOLOGY

The simulation twin's development involved four key stages: **Observation**, **Reconstruction**, **Simulation**, and **Appraisal**. These development steps crystallized into 2 final outputs:

- A playable simulation prototype with tools to track, input, and export session-based **configuration inputs** and **spin outcomes** · <http://lucky888.ludoai.xyz/>
- A dashboard with file upload functions and structured reporting · <http://lucky888.ludoai.xyz/dashboard/>

A. OBSERVATION

- Specific rules, mechanics and hidden parameters such as the minimum number of matching symbols required for a win were directly replicated from observable gameplay.
- Values such as internal symbol probabilities, exact feature-entry weights, and multiplier probabilities cannot be verified simply by observing the game. All parameters that cannot be verified have still been implemented with easily modifiable weights for iterative testing.

SYMBOL WEIGHTS AND PAYOUT DATA				
Symbol	Weight	minCount	maxCount	Payout
L1 (Low-1)	180 (16.38%)	8	9	0.25×
		10	11	0.75×
		12	30	2×
L2 (Low-2)	170 (15.47%)	8	9	0.4×
		10	11	0.9×
		12	30	4×
L3 (Low-3)	160 (14.56%)	8	9	0.5×
		10	11	1×
		12	30	5×
L4 (Low-4)	150 (13.65%)	8	9	0.8×
		10	11	1.2×
		12	30	8×
L5 (Low-5)	140 (12.74%)	8	9	1×
		10	11	1.5×
		12	30	10×
H1 (High-1)	90 (8.19%)	8	9	1.5×
		10	11	2×
		12	30	12×
H2 (High-2)	75 (6.82%)	8	9	2×
		10	11	5×
		12	30	15×
H3 (High-3)	60 (5.46%)	8	9	2.5×
		10	11	10×
		12	30	25×
H4 (High-4)	45 (4.09%)	8	9	10×
		10	11	25×
		12	30	50×
SCATTER	24 (2.18%)	N/A	N/A	N/A
MULTIPLIER	5 (0.45%)	N/A	N/A	N/A

MULTIPLIER WEIGHTS	
Multiplier	Weight
2×	1000 (18.27%)
3×	900 (16.44%)
4×	800 (14.61%)
5×	700 (12.79%)
6×	600 (10.96%)
8×	450 (8.22%)
10×	350 (6.39%)
12×	250 (4.57%)
15×	180 (3.29%)
20×	120 (2.19%)
25×	80 (1.46%)
50×	30 (0.55%)
100×	10 (0.18%)
250×	3 (0.055%)
500	1 (0.018%)

- Reel mechanics specific to the Bathala Slot machine game were carefully observed before replication. This is crucial because the game does not use paylines and instead relies on **tumbling logic**.

TUMBLE MECHANIC	BATHALA SKILL (SYMBOL ELIMINATION)
• After a win, winning symbols are removed. Remaining symbols fall into place, and new symbols drop in from above. • The grid is then checked again for another win. Tumbles continue until no new win is formed.	• After each winning <i>Tumble</i>, one Low Pay Symbol (L1 to L5) is selected and removed from the reels. • Removal does not award a payout by itself. • Remaining symbols fall into place, new symbols drop in, and the next Tumble is evaluated.
SCATTERS	FREE SPINS
• Scatter Symbols pay anywhere on the reel. ▪ 4 Scatters: 3× Total Bet ▪ 5 Scatters: 5× Total Bet ▪ >6 Scatters: 100× Total Bet • Landing ≥4 Scatters in the Base Game awards 15 Free Spins. • Note: Scatter Symbols are evaluated after all Tumbles have finished.	• Landing ≥4 Scatters in the Base Game awards 15 Free Spins. • Landing ≥3 Scatters <i>during</i> Free Spins awards 5 additional Free Spins. • Multipliers collected from winning Tumbles are added together and carried forward for the remainder of the Free Spins feature.
MULTIPLIERS	MAX WIN
• When a winning Tumble contains one or more Multiplier Symbols, their values are added together and applied to the win. • During Free Spins, collected multipliers are added to the current feature multiplier and accumulate into subsequent tumbles. • Note: Multiplier Symbols are collected only when they appear on a winning Tumble.	• Maximum Win: 10,000× Total Bet • Highest Multiplier Symbol: 500×

B. RECONSTRUCTION & SIMULATION

- Observable gameplay was translated into a simulation engine that could reproduce Spin behaviours.
- The simulation engine was designed to trigger the known reel logic: **symbol generation**, **win evaluation**, **multiplier handling**, **symbol eliminations**, **symbol refill**, **Scatter evaluation**, and **Free Game resolution**.

SYMBOL SET

L1
LOW

L2
LOW

L3
LOW

L4
LOW

L5
LOW

H1
HIGH

H2
HIGH

H3
HIGH

H4
HIGH

SCATTER
SPECIAL

×2
SPECIAL

GAME BOARD (6×5 REEL)

L1

L4

H3

L3

H1

H4

H1

SCATTER

L5

L3

L4

H1

H3

L3

H2

L4

L4

L2

L2

L4

L5

L1

H3

L1

L2

L3

H1

H1

L3

L2

SYMBOL REFILL TUMBLE LOGIC

L2

L1

L3

L3

L4

L5

H1

H3

H1

L1

L3

H4

H3

L5

L1

L3

L1

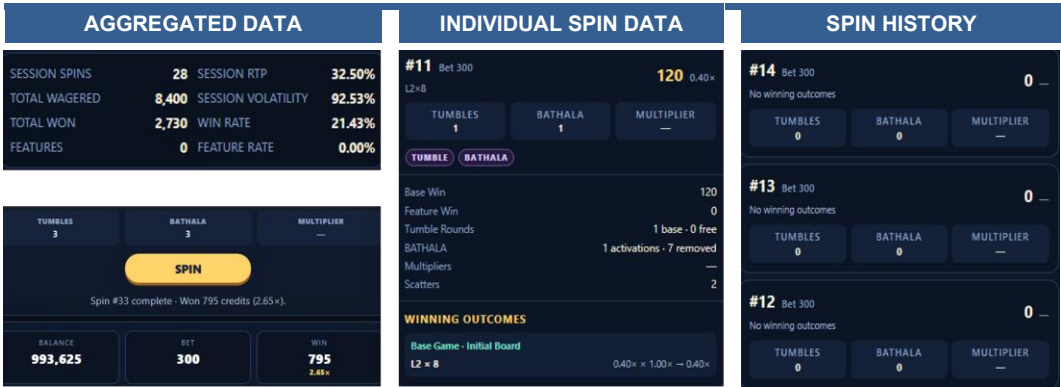
L1

L1

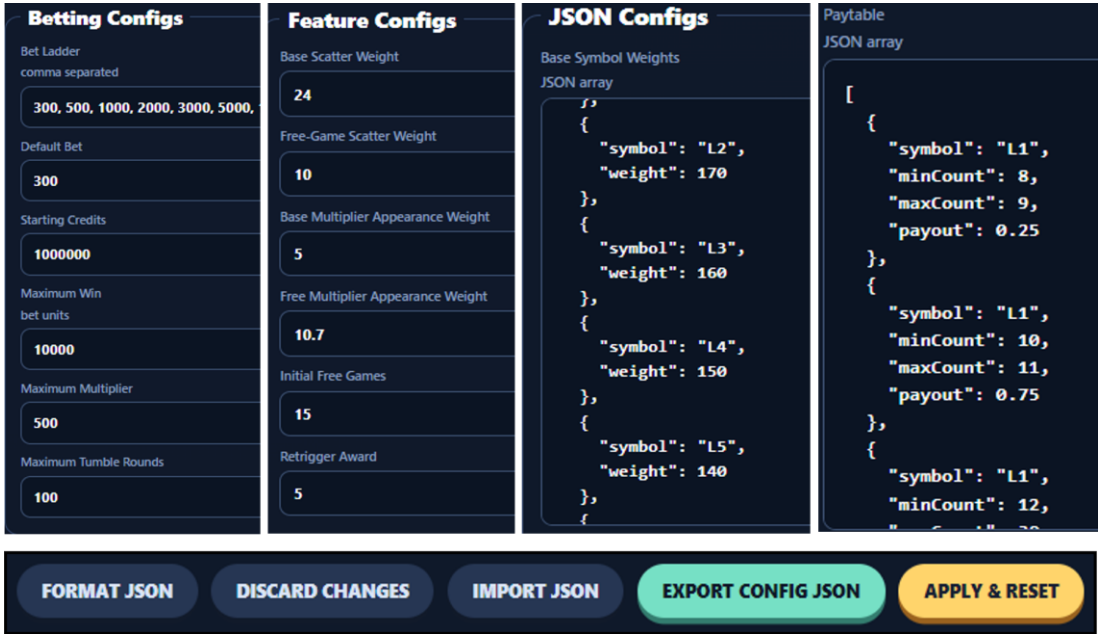
L5



- The simulation prototype updates session metrics in real-time whilst allowing each spin's inputs and outputs to be inspected independently.



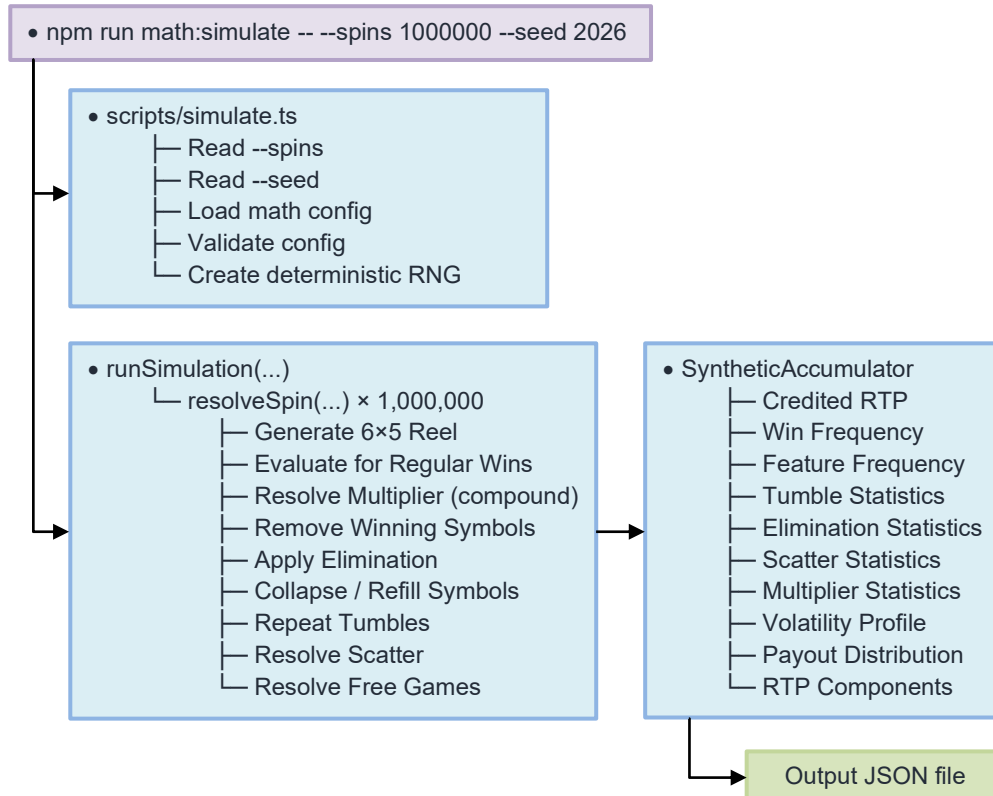
- Betting configs, feature configs, symbol weights and paytables can be modified, imported, tested and exported on a per session basis.



C. APPRAISAL

- To evaluate the game's mathematical profile, a standalone Monte Carlo simulation script was developed to generate large volumes of synthetic spin data.

- Deterministic **random seeds** allow results to be reproduced and validated at-scale, thereby providing greater confidence in the observed RTP, volatility, win rate, and feature performance.



- In order for the simulation data to be successfully evaluated, a math dashboard with a **file ingestion pipeline** was created to process the aggregated Spin data (JSON or CSV) into **performance metrics**.

FILE INGESTION WORKSPACE

Simulation Sets

Sim 1	Sim 2
ACTIVE Sim 1	ACTIVE Sim 2
Configuration: lucky888-bathala-aligned-v3	Configuration: lucky888-bathala-aligned-v3
Game Version: 3.0.0	Game Version: 2.0.0
Source file: lucky888-spin-history_lucky888-bathala-aligned...	Source file: bathala-simulation-2026-1000000.json
session spins: 1,000	Simulation Spins: 1,000,000
Session RTP: 171.81%	Credited RTP: 93.15%
Generated Timestamp: Aug 19, 2026, 9:12 PM	Generated Timestamp: Aug 19, 2026, 3:14 PM
Validation state: WORKBENCH CSV	Validation state: VALID REPORT
Version mismatch	Version mismatch
LOAD BUNDLED REPORT	LOAD BUNDLED REPORT
Choose bundled report	Choose bundled report
Replace Analysis File or browse file	Replace Analysis File or browse file
View Detailed Report	View Detailed Report
Remove Report	Remove Report
Report loaded into Sim 1	Report loaded into Sim 2

DASHBOARD SUMMARY

CREDITED RTP	FEATURE FREQUENCY	WINNING SPIN FREQUENCY	MAXIMUM OBSERVED WIN	COEFFICIENT OF VARIATION	FEATURE RTP CONTRIBUTION
93.15%	1 in 113	31.31%	3,821×	9.50	27.03%
CORE SIMULATION PROFILE			BASE SPINS VS. FEATURES		
CORE METRIC		RESULT	METRIC		
Credited RTP		93.15%	RTP Contribution		66.12%
Winning Spin Frequency		31.31%	Tumble Trigger		27.03%
Feature Frequency		1 in 113	Avg Rounds / Trigger		1.64
Maximum Observed Win		3,821×	Maximum Tumble Depth		10
Base RTP Contribution		66.12%	Multiplier RTP		22.97%
Feature RTP Contribution		27.03%	Scatter RTP		0.015%
Total Multiplier RTP Contribution		49.60%			

RTP COMPOSITION

RTP COMPOSITION

RTP Contribution by Component

Component	RTP Contribution
Base Regular	34.99%
Base Scatter	4.50%
Base Multiplier	26.63%
Free Regular	4.05%
Free Scatter	0.015%
Free Multiplier	22.97%

WINNING MECHANIC	CREDIT PAYOUT	RTP CONTRIBUTION
Base Regular	349,900	34.99%
Base Scatter	44,999	4.50%
Base Multiplier	266,335	26.63%
Free Regular	40,507	4.05%
Free Scatter	148	0.015%
Free Multiplier	229,652	22.97%
BASE GAME TOTAL	661,233	66.12%
FREE GAME TOTAL	270,306	27.03%
TOTAL RTP	931,539	93.15%

RTP SOURCE MIX

Regular Symbols

39.04%

Scatter Activations

4.51%

Multipliers

49.60%

MECHANICS OVERVIEW

TUMBLE ACTIVATIONS		BATHALA ACTIVATIONS	
Tumble Rounds per Paid Spin		Bathala Activations	
Tumble Triggers per Paid Spin		Bathala Elimination → Next Win Activation	
Average Rounds per Trigger Resolution		Bathala Activations per Paid Spin	
Maximum Observed Depth		Symbols Eliminated per Paid Spin	
FREE GAME		FEATURE ACTIVATIONS	
Feature Trigger Count		Initially Awarded Free Games	
Feature Frequency		Total Retriggers	
Average Free Games Played		Average Retriggers per Feature	
Feature RTP Contribution		Average Ending Free Game Multiplier	
MULTIPLIERS		SCATTER ACTIVATIONS	
Multiplier Appearance per Paid Spin		Base Scatter	
Average Multiplier Value		Free Scatter	
Average Summed Multiplier per Multiplied Win		Feature Frequency	
Maximum Summed Multiplier		Total Retriggers	
Total Multiplier RTP Contribution		FEATURE LENGTHS	
		VOLATILITY PROFILE	
		Mean Win per Paid Spin	
		Variance	
		Standard Deviation	
		Coefficient of Variation	
		Maximum Observed Win	

2. SUMMARY OF OBSERVATIONS

HEADLINE FINDINGS

93.15% Credited RTP	31.31% Winning Spin Frequency (≈1 winning outcome per 3.2 paid spins)
0.882% Feature Frequency (≈1 per 113 paid spins)	53.24% Multiplier Payout as share of Credited RTP
3,821× Maximum Observed Win	91.42 - 94.89% ~3.47% RTP Confidence Width

A. CORE OBSERVATIONS

✚ Credited RTP: 93.15%

- The default configuration produced an overall RTP of 93.15%, which means the game operates within a commercially plausible range. However, how that return is distributed between ordinary wins, multipliers, Scatters, and Free Games matter more for user engagement and monetization.

✚ Winning Spin Frequency: 31.31% (≈1 in 3.2 paid spins)

- The default configuration provides relatively frequent positive feedback despite the game's high-volatility profile. The result suggests that volatility is being generated less through low hit frequency, and more through substantial variation in the size of winning outcomes.

✚ Feature Frequency: 0.88% (≈1 in 113 paid spins)

- Free Games contribute approximately 29% of the game's total RTP. So, while the majority of RTP still comes from the Base Game, the feature is large enough to influence the player's overall risk-and-reward experience.

✚ Multiplier RTP Contribution: 49.60% (or 53.24% of Credited RTP)

- Multipliers are the single largest source of returned value in the current configuration.
- This makes the game a multiplier-led experience where relatively small changes to multiplier frequency and payout weighting can substantially alter RTP and volatility.

B. MULTIPLIER FREQUENCY & RTP

✚ Free Game Multiplier RTP: 22.97% (out of 27.03% Feature RTP)

- Approximately 85% of Free Game RTP is generated by multiplier uplift. This indicates that Free Game is fundamentally an accumulation and multiplier-led feature, rather than a feature driven by additional spins and regular wins.

✚ Base Game Multiplier RTP: 26.63% (out of 66.12% Base Game RTP)

- Multipliers are not only important during Free Games. They account for roughly 40.2% of Base Game return as well. This means multiplier behaviour affects the entire game economy rather than functioning solely as a bonus-feature enhancement.

✚ Multiplier Frequency: 4.85% (≈1 in 20.6 paid spins)

- Multipliers appear relatively infrequently yet generate half of total RTP, demonstrating how leveraged the mechanic is. Basically, whenever multipliers interact with winning combinations and tumblers, they produce a disproportionately large payout effect.

✚ Mean Multiplier vs. Summed Multiplier vs. Maximum Observed : 6.60x vs. 8.44x vs. 522x

- At 6.60x, most multiplier interactions seem to occur at relatively modest values.

- However, the large gap between the average and maximum observed summed multiplier is characteristic of a design where multiplier events provide constant engagement while rare combinations create headline win potential.

C. SYMBOL ELIMINATION (BATHALA SKILL)

Elimination Conversion Rate: 38.78%

- Approximately 4 in 10 symbol eliminations result in the refilled board immediately producing a winning combination.
- This observation confirms that the elimination effect is not merely a visual event but is in fact a continuation mechanic by virtue of regularly extending winning sequences.

Elimination Activations: 56.74% (≈ 1.95 symbols eliminated per paid spin)

- The elimination effect changes the reel composition frequently, allowing for more variance.
- Even when elimination does not immediately create another win, the mechanic boosts the number of refills and winning frequency, making it a core mechanic rather than a bonus event.

D. TUMBLE MECHANIC

Tumble Frequency & Average Depth: 34.74% & 1.63 rounds

- While roughly 1/3 of paid spins produce a tumble, the average tumble sequence remains relatively short. These sequences serve to provide continuation without long cascades dominating gameplay.
- Regardless, the maximum observed tumble depth of 10 indicates that players do occasionally encounter extended sequences.

Base Game Tumble vs. Free Game Tumble: 30.98% vs. 27.77%

- Free Games do not appear to derive more RTP from either deeper or more frequent tumbles.
- Instead, the feature's additional RTP is primarily generated through the accumulating multiplier mechanic. This is useful because it clearly separates the role of the two mechanics: tumbles provide cadence, while multipliers provide feature and payout escalation.

E. VOLATILITY & PAYOUT

Payout Tail: 10 $\times$ 1 in 68 paid spins (or 1 in 21 winning spins)

- 95% of winning spins remain below 10 $\times$, while $\geq 100\times$ outcomes represent only about 0.34% of winning spins.
- Moreover, only 34 out of 1 million paid spins triggered $\geq 500\times$ outcomes.
- This means that ordinary wins serve to maintain session activity and user engagement, while the most memorable wins remain genuinely coveted.

Mean Win per Paid Spin vs. SD vs. CoV: 0.932 vs. 8.85 vs. 9.5

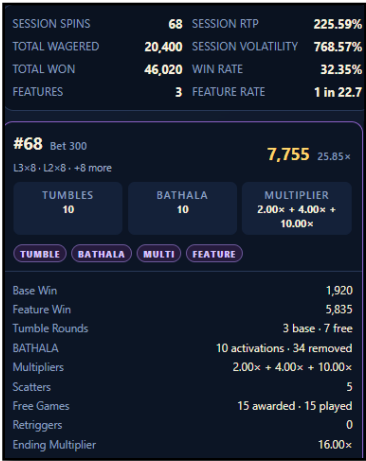
- The contrast between the mean win and a standard of deviation of 8.85 suggests that individual outcomes vary widely around the average.
- Since SD is approximately 9.5 times the mean RTP, a large proportion of RTP is concentrated on a handful of very high value wins rather than being distributed evenly. Therefore, this volatility profile is considered high.

3. EVALUATION & RTP TUNING

A. VOLATILITY PROFILE ANALYSIS

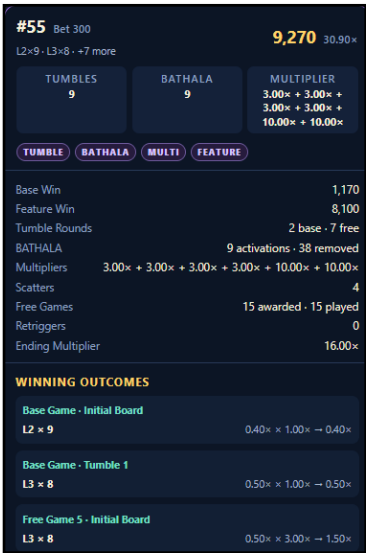
The current default setup in the simulation twin is a **multiplier-led** profile. In this setup, whilst some players will get to see more extravagant wins, the typical player will often only see outcomes around the average. This is further confirmed by a CoV of 9.5.

- From a qualitative experience standpoint, the 2 screenshots on the right show **Spins #68** and **#55** accumulate several multipliers, spiking session RTP to **225%** and volatility to a staggering **7.69**.
- One reason why the multipliers were able to accumulate so quickly was because of **3 feature activations** within 68 spins. Typically, this would be a very rare occurrence because the default config triggers 1 feature approximately every 120 paid spins.



Upon closer analysis of the exported CSV, over 80% of the session's total returns came from just 4 specific spins:

- Spin #40** (Free Game Trigger 1): **21.60x** total win
- Spin #44** (Base Game Outlier): **46.60x** total win
- Spin #55** (Free Game Trigger 2): **30.90x** total win
- Spin #68** (Free Game Trigger 3): **25.85x** total win



For a Slot machine and RMG operator, such variance is unsustainable.

- One way to resolve the volatility spikes is to replace the existing multiplier-led math profile with a **tumble-led math profile**, where engagement and RTP are primarily derived from symbol eliminations, longer tumble sequences, and medium-sized wins.
- The main KPI target is to reduce CoV to between **~4.5 and 6**.

PROFILE COMPARISON BY CHARACTERISTICS

Characteristic	Existing Profile (Multiplier-led)	Proposed Profile (Tumble-led)
Multipliers	Primary accelerator & Core mechanic	Occasional/Tertiary mechanic
Coefficient of Variation (CoV)	Current: ~7.5 to 9.5	Target: ~4.5 to 6
Small Wins	Frequent but relatively weak	More frequent with slightly larger wins
Medium Wins	Less frequent and relatively weak	Core winning mechanic
Extreme Wins	Key RTP contributor	Much rarer with lower caps
Free Games	Enhanced by Multiplier mechanic	Enhanced by premium Symbols
Engagement Design	Powerful Multiplier spikes	Frequent clusters and tumbles
Player Experience	"Nothing... nothing... BOOM"	Win → Tumble → Medium payouts

- To achieve this proposed profile, there are 4 key areas that must be tuned:

Tuning Area	Proposed Changes
Multiplier Weights	Move concentration from High Multipliers to Low Multipliers
Base Game Symbol Weights	Move concentration towards Low-Symbol distribution
Free Game Symbol Weights	Concentrate weightage towards premium Symbols (High-Symbols)
Cluster Paytable Payout Adjustments	Dampen the payout cliffs observed in High-Symbols, while raising the overall payouts for Low-Symbols

B. MULTIPLIER TUNING**DEFAULT MULTIPLIER STATISTICS**

6.57× Mean Multiplier	11.13× Multiplier Standard Deviation
4.46% ≥20× Frequency (≈1 per 22.4 paid spins)	0.80% ≥50× Frequency (≈1 per 125 paid spins)
0.26% ≥100× Frequency (≈1 per 384 paid spins)	0.073% ≥250× Frequency (≈1 per 1370 paid spins)

Firstly, the multiplier distribution is very dispersed and requires radical tuning. A mean multiplier of **4.5×** should be the hard cap, while multiplier SD should be below **3×**. This change requires dampening the existing lottery mechanic and turning it into a progression amplifier. The two tables below illustrate the scale and impact of the new configuration.

MULTIPLIER WEIGHTS

Multiplier	Current Config	New Weights
2×	1000 (18.27%)	1500 (23.09%)
3×	900 (16.44%)	1350 (20.79%)
4×	800 (14.61%)	1200 (18.48%)
5×	700 (12.79%)	950 (14.63%)
6×	600 (10.96%)	700 (10.78%)
8×	450 (8.22%)	400 (6.16%)
10×	350 (6.39%)	220 (3.39%)
12×	250 (4.57%)	100 (1.54%)
15×	180 (3.29%)	45 (0.69%)
20×	120 (2.19%)	20 (0.31%)
25×	80 (1.46%)	8 (0.12%)
50×	30 (0.55%)	2 (0.03%)
100×	10 (0.18%)	0 (0.00%)
250×	3 (0.055%)	0 (0.00%)
500×	1 (0.018%)	0 (0.00%)

MULTIPLIER STATISTICS

Observed Statistic	Current Config	Multiplier Targets
Mean Multiplier	6.57×	4.43×
Multiplier SD	11.13×	2.75×
≥10× Frequency	18.70%	6.08%
≥20× Frequency	4.46%	0.46%
≥50× Frequency	0.80%	0.03%
≥100× Frequency	0.26%	0.00%

While 100×, 250× and 500× have been made redundant, lower volatility does not mean more boring gameplay. The changes to the multiplier curve simply replaced volatility with event frequency.

- So, while ≥10× multipliers have decreased very significantly, multipliers between 2× and 6× have increased by a noticeable margin themselves.
- So statistically speaking, even though the major payouts have become even more rare, players will experience the lower multiplier events more frequently than ever before.

C. PAYTABLE TUNING

The default volatility profile has a fairly linear frequency curve for the 9 regular Symbols. However, this game is special in the sense that the regular Symbols are split into 2 distinct groups: **5 Low Symbols**, and **4 High Symbols**. This

grouping matters because of the Bathala Symbol Elimination Skill, where a randomly selected Low Symbol is eliminated after a regular Symbol Win. High Symbols are not able to benefit from this elimination.

- Due of this exclusion, Low Symbols actually add more to engagement than High Symbols, simply because Eliminations from Low Symbols can trigger recursive wins.
- Therefore, one way to noticeably reduce volatility while maintaining engagement is to shift both **symbol frequency weightage** and **payouts** from **High Symbols (H1→H4)** to **Low Symbols (L1→L5)**.

NEW PAYTABLE PAYOUTS

Base & Free Symbol	minCount	maxCount	Current Payout	New Payout
L1 (Low-1)	8	9	0.25×	(+60%) 0.4×
	10	11	0.75×	(+7%) 0.8×
	12	30	2×	(-25%) 1.5×
L2 (Low-2)	8	9	0.4×	(+25%) 0.5×
	10	11	0.9×	(+11%) 1×
	12	30	4×	(-50%) 2×
L3 (Low-3)	8	9	0.5×	(+20%) 0.6×
	10	11	1×	(+20%) 1.2×
	12	30	5×	(-50%) 2.5×
L4 (Low-4)	8	9	0.8×	(-6.25%) 0.75×
	10	11	1.2×	(+25%) 1.5×
	12	30	8×	(-59%) 3.25×
L5 (Low-5)	8	9	1×	(-10%) 0.9×
	10	11	1.5×	(+20%) 1.8×
	12	30	10×	(-60%) 4×
H1 (High-1)	8	9	1.5×	(-60%) 1.25×
	10	11	2×	(+25%) 2.5×
	12	30	12×	(-58%) 5×
H2 (High-2)	8	9	2×	(-25%) 1.5×
	10	11	5×	(-40%) 3×
	12	30	15×	(-53%) 7×
H3 (High-3)	8	9	2.5×	(-20%) 2×
	10	11	10×	(-55%) 4.5×
	12	30	25×	(-60%) 10×
H4 (High-4)	8	9	10×	(-70%) 3×
	10	11	25×	(-72%) 7×
	12	30	50×	(-68%) 16×

- As a key point of note, the default **H3** has a huge payout cliff. At (**H3 x 8**), the payout is **2.5×** which then scales rapidly to **10×** at (**H3 x 10**), and then **25×** at (**H3 x 12**). For a basic mechanic, such exponential payouts are unnecessary. Therefore, my suggested payable has reduced these to **2.5×**, **4.5×** and **10×** respectively.
- Similarly, the default **L5** went from a payout of only **1.5×** at (**L5 x 10**) to **10×** at (**L5 x 12**). I have suggested reducing (**L5 x 12**) from **10×** to just **4×**.
- In fact, I have compressed most of the payout cliffs from **L3**, **L4**, **L5**, **H1**, **H2**, **H3**, **H4** which will help us to free up payout allocations to **L1** and **L2**. So for example, the suggested payout at (**L1 x 8**) should increase by 60%. Performing these adjustments will materially reduce standard deviation.

D. SYMBOL WEIGHTAGES

In conjunction with higher payouts for Low Symbols, we can also improve their weight frequency. This will noticeably improve player experience because players will be able to see more 8-9 Symbol Wins, which could then lead to more **Tumbles**, more **Symbol Eliminations**, more **Symbol Refills**, and *recursive wins* thereafter.

In fact, the frequency weights for **Base Game Symbols** and **Free Game Symbols** should be modified independently from one another as they impact RTP differently.

NEW BASE GAME SYMBOL WEIGHTS			NEW FREE GAME SYMBOL WEIGHTS		
Base Game Symbols	Current Config	New Weights	Free Game Symbols	Current Config	New Weights
L1	180 (16.38%)	(+30) 210 (19.34%)	L1	170 (16.03%)	(+10) 180 (16.30%)
L2	170 (15.47%)	(+25) 195 (17.78%)	L2	160 (15.08%)	(+5) 165 (14.95%)
L3	160 (14.56%)	(+10) 170 (15.50%)	L3	150 (14.14%)	(-5) 145 (13.13%)
L4	150 (13.65%)	(-5) 145 (13.22%)	L4	140 (13.20%)	(-20) 120 (10.87%)
L5	140 (12.74%)	(-20) 120 (10.94%)	L5	130 (12.26%)	(-30) 100 (9.06%)
H1	90 (8.19%)	(-10) 80 (7.29%)	H1	95 (8.96%)	(+55) 150 (13.59%)
H2	75 (6.82%)	(-10) 65 (5.93%)	H2	80 (7.54%)	(+40) 120 (10.87%)
H3	60 (5.46%)	(-10) 50 (4.56%)	H3	65 (6.13%)	(+5) 70 (6.34%)
H4	45 (4.09%)	(-10) 35 (3.19%)	H4	50 (4.71%)	(-10) 40 (3.62%)
SCATTER	24 (2.18%)	24 (2.19%)	SCATTER	10 (0.94%)	10 (0.91%)
MULTIPLIER	5 (0.45%)	(-2) 3 (0.27%)	MULTIPLIER	10.7 (1.01%)	(-6.7) 4 (0.36%)

For **Base Game Symbols**, I have intentionally shifted weightage from **L4**, **L5**, **H1**, **H2**, **H3** and **H4** in favor of **L1**, **L2** and **L3**. Doing so increases the probability of triggering at least one ≥ 8 regular Symbol cluster from **~32%** to **~49%** using this distribution method.

- At the same time, I have reduced the frequency of multipliers by about **40%** (from **0.45%** to **0.27%**).
- So as a result, instead of waiting for a multiplier event for engagement, the player will now expect:
 ≥ 8 REGULAR SYMBOL $\rightarrow$ BATHALA ELIMINATION $\rightarrow$ REFILL $\rightarrow$ ≥ 8 REGULAR SYMBOL REACTIVATION

On the other hand, I would recommend making **Free Game Symbol** weightage premium-leaning. So for example, increase **H1** from **9.0%** to **13.6%**, while increasing **H2** from **7.5%** to **10.9%**.

- This might create an interesting psychological effect where **Free Games** become perceived as a premium variant due to the frequency of High-Symbols.

E. FINAL EVALUATION

Targeted Metrics			
Performance Metric	Sim1 (Multiplier-led)	Sim2 (Tumble-led)	Delta
Win Cadence	31.3%	45.6%	+14.3%
Tumbles	34.7%	49.1%	+14.4%
Multiplier RTP	49.6%	24.4%	-25.2%
Volatility (CoV)	9.50	3.68	-61.2%
Max Observed Win	3,820×	616×	-83.9%

- Sim2 (Tumble-led) delivers a much **more active core loop**.

- Win Cadence rose from **31.3%** to **45.6%** and Tumbles increased from **34.7%** to **49.1%**, meaning players encountered continuation mechanics far more frequently.

✚ Sim2 (Tumble-led) is significantly **less multiplier-dependent**.

- Multiplier RTP fell from **49.6%** to **24.4%**, so Sim2 shifts more of its payout value towards regular and tumble-driven outcomes instead of multiplier spikes.

✚ Sim2 (Tumble-led)'s **volatility is dramatically compressed**.

- CoV fell from **9.50** to **3.68**, with Maximum Observed Win falling from **3,820×** to **616×**. This indicates a much narrower payout distribution, with fewer extreme tail outcomes and a more consistent player experience.

Performance Metric	Sim1 (Multiplier-led)	Sim2 (Tumble-led)	Delta
Simulation Overview			
Simulation Spins	1,000,000	1,000,000	N/A
Credited RTP	93.15%	94.00%	+0.85%
95% RTP Confidence	91.4% to 94.9%	93.3% to 94.7%	N/A
Winning Spin Frequency	31.31%	45.59%	+14.3%
Feature Frequency	1 in 113	1 in 111	+1.9%
Tumbles & Bathala Elimination Skill			
Tumble Frequency	34.74%	49.12%	+14.4%
Tumble Depth (per spin)	0.567	0.91	+60.3%
Tumble Depth (per trigger)	1.633	1.852	+13.4%
Symbol Elimination → Win	38.78%	46.01%	+7.2%
RTP Composition			
Base RTP Contribution	66.12%	82.53%	+16.4%
Feature RTP Contribution	27.03%	11.47%	-15.6%
Base Regular RTP	34.99%	59.60%	+24.6%
Base Multiplier RTP	26.63%	18.07%	-8.6%
Free Game Regular RTP	4.05%	5.12%	+1.1%
Free Game Multiplier RTP	22.97%	6.33%	-16.6%
Total Multiplier RTP	49.60%	24.40%	-25.2%
Volatility & Payout Tail			
Coefficient of Variation	9.50	3.68	-61.2%
Standard Deviation	8.85	3.46	-60.9%
Average Win / Winning Spin	2.98×	2.06×	-30.7%
Average Multiplier Value	6.60×	4.42×	-33.1%
Max Summed Multiplier	522×	55×	-89.5%
Maximum Observed Win	3,820×	616×	-83.9%
≥100× Occurrence	1 in 1,080	1 in 3,509	-69.2%
≥250× Occurrence	1 in 8,333	1 in 250,000	-96.7%
≥500× Occurrence	1 in 41,667	1 in 1,000,000	-95.8%
≥1,000× Occurrence	1 in 125,000	Nil	N/A